Assignment - 11. Sparsha Rajanna.

Write down, including the calculation procedure

1) Write the representation of 4 bit code 1101 into 7-bit, even parity Hamming code.

Sol^n parity bits 1, 2, 4 $(2^0, 2^1, 2^2)$

data bits 3, 5, 6, 7

4-bit data 1101 is positioned at 3, 5, 6, 7

$d_3 = 1$, $d_5 = 1$, $d_6 = 0$, $d_7 = 1$

initial arrangement.

$P_1, P_2, d_3, P_4, d_5, d_6, d_7$

$P_1, P_2, 1, P_4, 1, 0, 1$

001 → P_1
010 P_2
011 P_3
100 P_4
101 P_5
110 P_6
111 P_7

Calculating parity bits

P_1 covers positions 1, 3, 5, 7

$P_1 \oplus d_3 \oplus d_5 \oplus d_7 = 0$

$P_1 \oplus 1 \oplus 1 \oplus 1 = 0$

$\boxed{P_1 = 1}$

P_2 covers positions 2, 3, 6, 7

$P_2 \oplus d_3 \oplus d_6 \oplus d_7 = 0$

$P_2 \oplus 1 \oplus 0 \oplus d_7 = 0$

$\boxed{P_2 = 0}$

P_4: covers positions 4, 5, 6, 7

$$P_4 \oplus d_5 \oplus d_6 \oplus d_7 = 0$$

$$P_4 \oplus 1 \oplus 0 \oplus 1 = 0$$

$$\boxed{P_4 = 0}$$

The 7-bit Hamming code is

$$P_1\ P_2, d_3, P_4, d_5, d_6, d_7 = 1, 0, 1, 0, 1, 0, 1$$

Hamming code : 1010101

2) We have input 1101 which was encoded as 0011001
Find error position?

Soln Received code is

$P_1 = 0$, $P_2 = 0$, $d_3 = 1$, $P_4 = 1$, $d_5 = 0$, $d_6 = 0$, $d_7 = 1$.

Verifying parity conditions:

P_1: covers positions 1, 3, 5, 7

$$P_1 \oplus d_3 \oplus d_5 \oplus d_7 = 0$$

$0 \oplus 1 \oplus 0 \oplus 1 = 0$ [P_1 is OK]

P_2: covers positions 2, 3, 6, 7

$$P_2 \oplus d_3 \oplus d_6 \oplus d_7 = 0$$

$0 \oplus 1 \oplus 0 \oplus 1 = 0$ [P_2 is OK]

P_4: covers positions 4, 5, 6, 7

$$P_4 \oplus d_5 \oplus d_6 \oplus d_7 = 0$$

$1 \oplus 0 \oplus 0 \oplus 1 = 0$ [P_4 is OK]

Syndrome vector is calculated using the parity bit results in reverse order. (P_4, P_2, P_1)

Syndrome = $[P_4, P_2, P_1] = [0, 0, 0]$

Syndrome vector of 000 indicates no detected error in the given Hamming code 0011001.

But the input 1101 cannot be encoded as 0011001 because the position of the data bits in a Hamming code follow a specific structure & 0011001 does not adhere to this structure for the given input

Correct Hamming code for input 1101 is

1010101. $(d_3=1, d_5=1, d_6=0, d_7=1)$

The given encoded 0011001 cannot represent 1101

$d_3=1$ $d_5=0$, $d_6=0$, $d_7=1$

These data bits are not equal to 1101 $(d_3=1, d_5=1, d_6=0, d_7=1)$

It represents a different input not 1101.

By comparing given code 0011001 to correct code 1010101 bit by bit

The mismatches occurs at positions 1, 4 and 5